

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

INDIA FINANCIAL ANALYTICS USING POWER BI

A Mini Project Report

Submitted by

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B.Tech. Artificial Intelligence and Machine Learning

AD23632 - Framework for Data and Visual Analytics

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Project at a glance

This project turns a compact, real-world India macro-financial dataset into an offline, interactive Power BI report. Users can compare long-run trends, inspect the imported records, and narrow the view with year slicers while retaining the source data.

175	7	25 years	4 pages
indicator-year rows	annual series	2000-2024	in the PBIX

Project objective

Build a complete Power BI Desktop analytics application that presents Indian financial access, credit, inflation, interest-rate, money-supply and growth indicators in a clear report. Demonstrate source-data inspection, Power Query preparation, interactive filtering, and responsible interpretation of missing observations.

#	Goal	What the project does
		Import and type the source fields, preserve missing values, and make the indicator/year grain explicit.
02	Explore trends	Compare each indicator over time using its published unit and the latest available year.
		Show how a 2020-2024 year selection changes trend context and visible records.
04	Share offline	Package the report and imported model in a single Power BI Desktop file.

Deliverable

The native PBIX is the main application file. It contains four report pages: Financial overview, Cleaned data & slicers, Raw imported data, and Sources & presentation. The project also records its data source, transformation approach, definitions, and limitations.

Dataset and scope

The report uses a locally stored snapshot of annual India observations from seven World Bank indicator series. Every series is represented once per year from 2000 through 2024, giving a consistent 25-year row structure; values that were unavailable in the snapshot are blank rather than imputed.

Indicator code	Series	Unit	Observed	Latest available
FB.ATM.TOTL.P5	ATMs	per 100,000 adults	20 / 25	2024 | 24.47
FB.CBK.BRCH.P5	Commercial bank branches	per 100,000 adults	21 / 25	2024 | 14.62
FM.LBL.BMNY.GD.ZS	Broad money	% of GDP	22 / 25	2021 | 82.10
FP.CPI.TOTL.ZG	Consumer price inflation	%	25 / 25	2024 | 4.95
FR.INR.LEND	Lending interest rate	%	23 / 25	2022 | 8.57
FS.AST.PRVT.GD.ZS	Private sector credit	% of GDP	25 / 25	2024 | 41.61
NY.GDP.MKTP.KD.ZG	Real GDP growth	%	25 / 25	2024 | 7.10

DATA CHECK	RESULT	INTERPRETATION
Rows and keys	175 rows; 7 indicators x 25 years	One indicator-year row per series-year in the snapshot.
Observed values	161 numeric; 14 blank	Blank means no reported value in the retrieved series; it is not zero.
2020-2024 window	35 rows; 30 observed; 5 blank	Filtering narrows the five-year view without deleting underlying rows.

Source and license

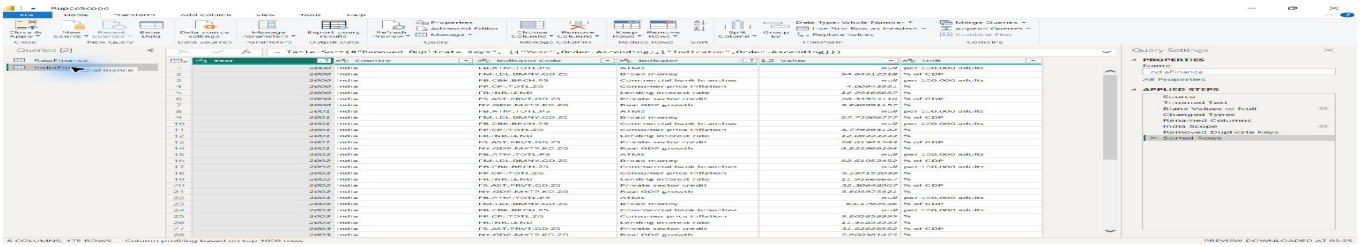
World Bank World Development Indicators (WDI), India country code IND. Indicator metadata identifies differing original sources across series; for example, the lending-rate series is sourced from the IMF's International Financial Statistics database. Snapshot retrieved 8 October 2026. Dataset attribution follows CC BY 4.0; see the linked source pages and license in the project materials.

World Bank - India data | Lending-rate source metadata | CC BY 4.0

Data preparation in Power Query

A separate raw import is retained for comparison. The prepared IndiaFinance query applies a transparent sequence of type and labeling steps to make the data usable in the report while preserving each indicator's unit and missing-value meaning.

STEP	APPLIED STEP	PURPOSE
01	Source	Read the embedded CSV snapshot into the model.
02	Trimmed Text	Normalize text labels and remove surrounding whitespace.
03	Blank Values to Null	Represent empty published observations as nulls.
04	Changed Types	Set Year to whole number and Value to decimal.
05	Renamed Columns	Use readable field names for reporting.
06	India Scope	Keep the project country scope explicit.
07	Removed Duplicate Keys / Sorted Rows	Keep the indicator-year key unique and order the result for review.



Power Query Editor evidence: prepared fields and Applied Steps shown in the project capture.

Inspecting the imported records

The Raw Imported Data page retains the original field names and the imported table. It provides a useful audit point before transformations: source codes, labels, years, values, units, and blank cells can be reviewed together.



Power BI Desktop - raw CSV import, showing original columns and the 175-row input.

Model grain

Each record describes one indicator for one year in India. The year is not a transaction date and the rows are not individual bank accounts. Values therefore need to be read within their indicator series and unit.

Measure design

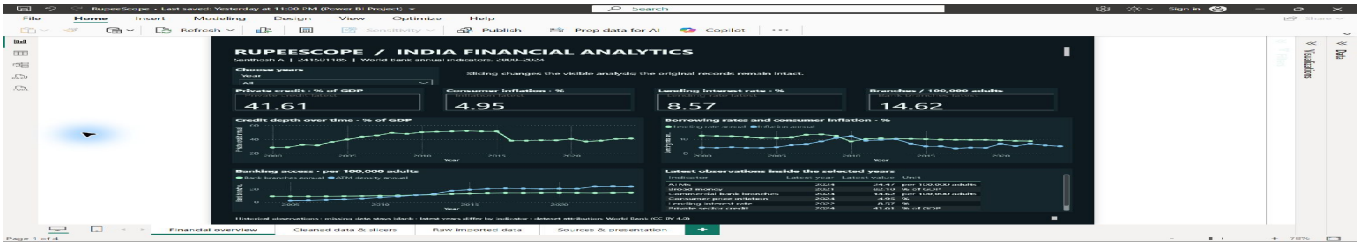
The model uses measures for observed and missing counts and calculates a latest value only for a single selected indicator. It does not add together rates, percentages of GDP, or banking-density values. For a selected year window, a trend or average should be interpreted only within the corresponding series.

Raw values preserved

The Power Query workflow changes types and presentation labels while retaining the source observations. This keeps the project suitable for demonstrating the difference between raw and prepared data without fabricating errors or filling missing values with zero.

Dashboard view: full historical context

With the year slicer cleared, the financial overview shows all available years in the snapshot. KPI cards display the latest non-missing observation per indicator, while line charts show the year-by-year patterns and a latest-observation table makes coverage differences visible.

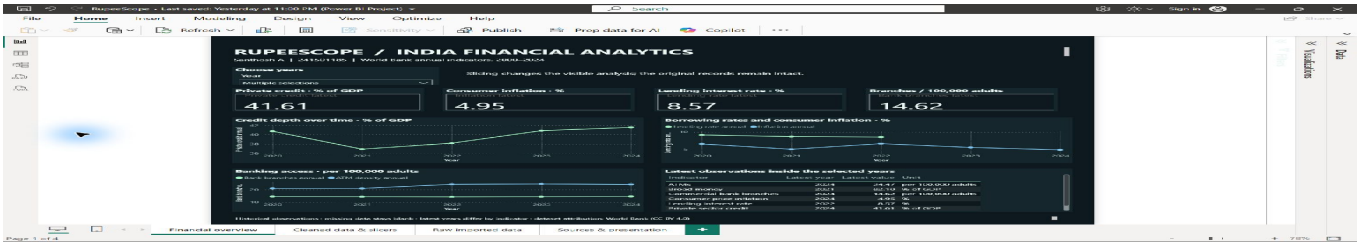


Baseline overview in Power BI Desktop: full 2000-2024 timeline with year slicer set to All.

VISUAL	QUESTION IT HELPS ANSWER
KPI cards	What are the latest available values for credit, inflation, lending rate, and bank branches?
Credit trend	How did private credit as a share of GDP vary through the snapshot period?
Rates and inflation	How do the available lending-rate and inflation series move over time?
Banking access	How did ATMs and commercial branches per 100,000 adults develop?

Dashboard view: selected years

Selecting 2020-2024 narrows the timeline from 25 years to five years. The report's year filter keeps 35 indicator-year rows in view: 30 values are reported and five are blank. Latest-value cards can remain unchanged when the latest available observation is still inside the selected range.



The same overview after selecting 2020-2024. The charts redraw on the selected year range; the KPI values remain anchored to each indicator's latest available observation within that selection.

READ THE FILTER CAREFULLY

A slicer changes the visible analytical context; it does not erase the imported source records. Series also have different end years, so a latest value shown on a card may be from an earlier year than another card.

Before and after summary

Baseline: 175 rows across 2000-2024, with 161 observed values. Filtered 2020-2024: 35 rows, 30 observed values, and 5 blanks. The filtered dashboard uses the same report page and visual layout so that the change in time context is easy to compare.

Findings, interpretation and limitations

The dashboard is a descriptive view of historical Indian financial and macroeconomic indicators. The snapshot illustrates the breadth of coverage and how banking access, inflation, credit and growth can be explored together while respecting each series' unit and data availability.

SNAPSHOT OBSERVATION	LATEST VALUE IN THIS FILE	READING NOTE
Private credit	2024: 41.61 % of GDP	Latest year differs by series; compare within its own timeline.
Consumer price inflation	2024: 4.95 %	Latest year differs by series; compare within its own timeline.
Real GDP growth	2024: 7.10 %	Latest year differs by series; compare within its own timeline.
ATM density	2024: 24.47 per 100,000 adults	Latest year differs by series; compare within its own timeline.
Lending interest rate	2022: 8.57 %	Latest year differs by series; compare within its own timeline.
Broad money	2021: 82.10 % of GDP	Latest year differs by series; compare within its own timeline.

Limitations

• The included CSV is a dated snapshot, not a live feed; later World Bank revisions may differ.
• Indicator coverage and latest available years vary. Broad money ends in 2021 and lending rate ends in 2022 in this snapshot.
• The dashboard supports description and comparison; it does not prove causation, provide forecasts, or recommend financial decisions.
• Annual aggregate data can hide regional, household, sectoral, and within-year variation.

Open and present

Open Santhosh-A-Financial-Analytics.pbix in Power BI Desktop. The dataset and Power Query source are embedded in the PBIX, so the report is designed to open without reconnecting to online services or the original computer's file paths. Use the Financial overview page, clear its year slicer, then choose 2020-2024 to reproduce the comparison. Use the Raw imported data and Cleaned data & slicers pages to explain the preparation flow.

Sources

World Bank, World Development Indicators, India country page: https://data.worldbank.org/country/india
World Bank indicator metadata, lending interest rate (FR.INR.LEND), original source listed as IMF International Financial Statistics: https://data.worldbank.org/indicator/FR.INR.LEND?locations=IN
World Bank indicator data pages: FP.CPI.TOTL.ZG (inflation), FS.AST.PRVT.GD.ZS (private credit), NY.GDP.MKTP.KD.ZG (GDP growth), FB.CBK.BRCH.P5 (bank branches), FB.ATM.TOTL.P5 (ATMs), FM.LBL.BMNY.GD.ZS (broad money).
Data license: Creative Commons Attribution 4.0 International (CC BY 4.0), https://creativecommons.org/licenses/by/4.0/. Project dataset snapshot retrieved 8 October 2026.

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